

1. Administrative & Study Identification

Full Study Title: A Phase 1 Study of R07623066 Alone and in Combination in Patients With Advanced Solid Tumors **Short Title/ Acronym:** R07623066 (KSQ-4279) Phase 1 Study **Protocol Number:** WP45169 / KSQ-4279-1101 **NCT Number:** NCT05240898 **Posting ID:** 801086415073291405 **Sponsor/Company Name:** Roche / Genentech / KSQ Therapeutics, Inc. **Investigational Product:** R07623066 (KSQ-4279), a USP1 Inhibitor **Phase of Development:** Phase I **Trial Status:** Active, Not Recruiting (ACTIVE_NOT_RECRUITING) **Medical Monitor:** Genentech Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the safety, tolerability, and pharmacokinetics (PK) of R07623066 administered as a monotherapy or in combination with other agents, and to determine the Maximum Tolerated Dose (MTD) measured by the Incidence of Dose Limiting Toxicities (DLTs). * **Secondary Objectives:** To evaluate preliminary clinical activity including Objective Response Rate (ORR) per RECIST v1.1 Investigator assessment, incidence and severity of Treatment Emergent Adverse Events (TEAEs) and change from baseline in laboratory results, and to assess the PK parameters of R07623066.

2.2 Background & Rationale To address the need for targeted therapies in advanced solid tumors, particularly those with homologous recombination repair (HRR) mutations. KSQ-4279 (R07623066) is a first-in-class small molecule inhibitor of ubiquitin-specific protease 1 (USP1). Inhibiting USP1 in cancer cells limits their ability to repair DNA damage, suppressing tumor growth and sensitizing them to DNA-damaging therapies.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm (Dose Escalation and Dose Expansion cohorts) * **Blinding:** Open-label * **Randomization:** N/A

3.2 Planned Enrollment & Duration * **Target \$N\$:** 250 participants (Evaluated per standard Phase 1 dose-escalation rules, which includes a minimum of 12 participants specifically for the Food Effect cohort) * **Number of Sites:** Multicenter (United States) * **Estimated Study Start:** 25-Aug-2021 * **Estimated Primary Completion:** 25-Nov-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old. 2. Histologically or cytologically confirmed locally advanced (unresectable) or metastatic solid tumors with a life expectancy ≥ 12 weeks. 3. Measurable disease per RECIST v1.1 (required for expansion and backfill cohorts), ECOG performance status of 0 or 1, and adequate bone marrow function (absolute neutrophil count $\geq 1.5 \times 10^9/L$, platelet count $\geq 100.0 \times 10^9/L$, hemoglobin ≥ 10.0 g/dL). 4. Adequate renal (creatinine clearance ≥ 40 mL/min) and hepatic function.

4.2 Exclusion Criteria 1. Prior anticancer treatment including chemotherapy or small molecule-targeted therapy < 2 weeks prior to the first dose of study treatment. 2. Radiation therapy < 4 weeks prior to initiation of study treatment (stereotactic body radiation allowed < 2 weeks prior). 3. Requirement for ongoing treatment with strong or moderate CYP3A4 inhibitors or inducers.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: R07623066 administered orally once daily (QD) continuously in escalating dose levels. Also tested in combination with standard of care agents (Olaparib or carboplatin [Trade Name: Paraplatin, ATC Code: L01XA02]). *** Route of Administration:** Oral. *** Duration of Treatment:** Continues until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy * Permitted: Standard of care Olaparib or carboplatin for designated combination cohorts. *** Prohibited:** Strong or moderate CYP3A4 inhibitors or inducers; concurrent antibody therapy < 5 half-lives from the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline RECIST v1.1 Imaging, Labs
Baseline	Day 0	First Dose of R07623066, Baseline PK/PD Blood Draws
Interim	Continuous Cycles	Safety Labs, Intensive PK Profiling, Efficacy Assessment via RECIST v1.1
End of Study		
Follow-up	30-day safety follow-up post-last dose; Survival follow-up every 3 months for up to 1 year	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * Definition: The Maximum Tolerated Dose (MTD) Measured by the Incidence of Dose Limiting Toxicities

(DLTs). * **Measurement Method:** Clinical safety evaluation per NCI-CTCAE v5.0 and PK concentration analysis.

7.2 Secondary & Exploratory Endpoints * Preliminary anti-tumor activity including Objective Response Rate (ORR) per RECIST v1.1 Investigator assessment. * Incidence and severity of Treatment Emergent Adverse Events (TEAEs) and change from baseline in laboratory results. * Pharmacokinetics (PK) parameters of R07623066.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous clinical and laboratory monitoring throughout the dose escalation and expansion phases.
 - **Serious Adverse Events (SAEs):** Standard 24-hour reporting to the sponsor and institutional review boards (IRB).
 - **Data Monitoring Committee (DMC):** An internal Dose Escalation Review Committee reviews safety and PK data before approving escalation to the next dose cohort.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Set (all patients who receive at least one dose of study drug) and Efficacy-Evaluable Set (patients with baseline and post-baseline RECIST v1.1 scans).
 - **Sample Size Justification:** Traditional Phase 1 dose-escalation statistical model; the targeted enrollment of 250 is driven by dose-finding cohorts and safety expansion rather than formal power calculations for efficacy.
 - **Handling of Missing Data:** Standard handling for early-phase PK models and safety datasets.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine onsite and remote monitoring of clinical data, source documents, and RECIST v1.1 radiographic reads.
 - **Audit Trail:** Comprehensive eCRF locking and query resolution tracking for critical PK and dose-limiting toxicity data points.
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11. Study Identifiers & Governance

- **Primary ID:** WP45169 / KSQ-4279-1101

- **Registry Number:** NCT05240898
 - **Sponsor/Lead Organization:** Roche / Genentech / KSQ Therapeutics, Inc.
 - **Study Title:** A Phase 1 Study of R07623066 Alone and in Combination in Patients With Advanced Solid Tumors
 - **Official Regulatory Title:** A Phase 1 Study of R07623066 Alone and in Combination in Patients With Advanced Solid Tumors
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12. Clinical Framework (Oncology Specific)

- **Primary Condition / Disease Area:** Advanced Solid Tumors (Advanced Solid Malignancies)
 - **Condition Definition:** A malignant solid neoplasm that has spread extensively to other anatomic sites or is no longer responding to treatment.
 - **Preferred UMLS Name:** Solid Neoplasms, Advanced
 - **Preferred Name:** Advanced Malignant Solid Neoplasm
 - **Semantic Type:** Neoplastic Process
 - **SNOMED ID:** 39958004
 - **SNOMED Hierarchy:** 785317006|Advanced solid malignant neoplasm|Advanced solid malignant neoplasm (disorder)
 - **MeSH Code:** N/A
 - **Medical Methodology:** Targeted therapy via USP1 inhibition (DNA damage response pathway targeting)
 - **Optimization Target:** Dose escalation to define the MTD safely
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13. Study Procedures & Methodology

- **Management Pathway:** Stepwise dose escalation followed by dose expansion and dedicated combination/backfill cohorts
 - **Education Protocol (HCP):** Investigator protocol training emphasizing DLT definitions, PK sampling schedules, and NCI-CTCAE v5.0 grading.
 - **Education Protocol (Patient):** Informed consent detailing Phase 1 risks, intensive PK blood draw requirements, and food-effect cohort parameters.
 - **Quality Audit Mechanism:** Centralized radiographic reviews for RECIST v1.1 and stringent tracking of dose modification rules.
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14. Observation & Follow-Up Schedule

- **Total Duration:** Active treatment until progression/toxicity, plus up to 1 year of follow-up
- **Onsite Visit Cadence:** Highly frequent during Cycle 1 (Dose Escalation), transitioning to standard cycle-based visits.
- **Remote Monitoring Frequency:** Regular safety monitoring by study coordinators.

- **Checklist Review Interval:** Every 3 months during the 1-year post-treatment follow-up period

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				